



For the lulz

There are 10 types of people in the world: Those who understand binary, and those who don't.



New planets discovered!

Nasa has announced the single largest group of new validated planets, to date. 1284 new planets discovered, using the Kepler space telescope. <http://digit.in/NwPlnts>

Phone cams



All these photos were taken in studio conditions, with bright white lights and a white background. The Samsung Galaxy S7 (Pic 1) and iPhone 6S Plus (Pic 2), both manage the shadows well, but colours look more vivid from the Galaxy S7. The Moto G4 Plus (Pic 3), on the other hand, takes a good photo, considering the price.

goes down. Under low light, for example, outdoors in the evening, the phone more or less fails, as far as photography is concerned. There's too much noise and most of the details are lost. The phone has an ISOCELL sensor from Samsung, and while the image quality is satisfactory for a budget device, that's all there is to it. Under white lights, the white balance went off, turning a white background almost into blue.

It can take shots that are satisfactory, if not absolutely good. Images aren't very sharp, but the colours look fine. The main weakness of this camera is in reproducing details. Taking macro shots, of flowers, plants etc. doesn't work out very well.

While processing time and shutter lag aren't an issue, you can't always depend on the auto focus on this camera. This means that while taking quick shots, you'll often

find that the camera couldn't focus right, or hasn't exactly judged the scene very well. It's expected from a budget smartphone, but the Le 1s is definitely not the best camera phones we've seen in this range.

The Le 1s Eco runs on a MediaTek Helio X10 SoC, which gives it pretty good performance for multi tasking. That said, its performance, with respect to gaming, is nothing compared to the Redmi Note 3. While the

How We Tested

Over the years, we've seen many phones that can take good images, but have other, major weaknesses. The Sony Xperia Z5, for example, had one of the best cameras in its time, but it took too long to focus and shoot images. So, the first few things to test, included the focus time, processing and shutter delay, and the app launch times. You will barely find a good camera phone that doesn't get these parts right. You could find one in 2014, but not in the current generation.

Moving on, image quality is of course, the primary concern for anyone interested in a phone's camera. We took five common light conditions – bright sunlight, indoor tube lights, bright white studio lights, yellow lights, and the dreaded low light conditions. The goal was to represent your camera usage, for everything from an outdoor picnic, to an unevenly lit club. Only few shall pass all of these tests.

From here, we moved on to determine the specs and hardware used for the camera. This includes the type of sensor used, and the

aperture. To explain, a Back Side Illuminated (BSI) sensor, is usually preferred, but not always the only option. Samsung's ISOCELL technology is quite popular, and there are some other sensors that OEMs have used in the past.

At this point, you're probably wondering why we're not talking about the megapixel count. That's because while companies would have you believe otherwise, to benefit sales, it doesn't really play as big a role. Sure, higher megapixel counts can be great, but this test essentially proves that the best cameras don't always have the highest megapixel count. To be clear, the numbers did play a role in this comparison, but they weren't the determining factor. The most emphasis was given to the image quality, followed by the speed and other factors mentioned above, and then came the megapixel count.

So, with about 70 percent weightage being given to the camera, we moved on to determining the quality of the phones. This includes

testing performance, rating the looks, judging the build quality and a lot more. We ran the usual benchmarks, including, AnTuTu 6.0, Geekbench 3 and GFX Bench. While AnTuTu gives you a peek at the phone's overall system performance, GFXBench is a very graphic centric benchmark. Geekbench, on the other hand, is good at determining the single core and multi-core performance of these phones.

We'll be honest with you, it's an exhausting endeavour, especially when one accounts for how clumsy we humans can be. Nevertheless, all of these tests were then combined with real world performance and usage, to come to our final conclusions. sixteen phones entered the fray, and they all fought hard.

Our competitors include, the LeEco Le 1s Eco, Xiaomi Redmi Note 3, Moto G4 Plus, HTC Desire 828 Dual SIM, HTC Desire 728G, OnePlus X, Moto X Play, Lenovo Vibe Shot, Vivo V3 Max, Xiaomi Mi 5, HTC One A9, Oppo F1 Plus, LG Nexus 5X, Huawei Nexus 6P, Apple iPhone 6S Plus and the Samsung Galaxy S7.